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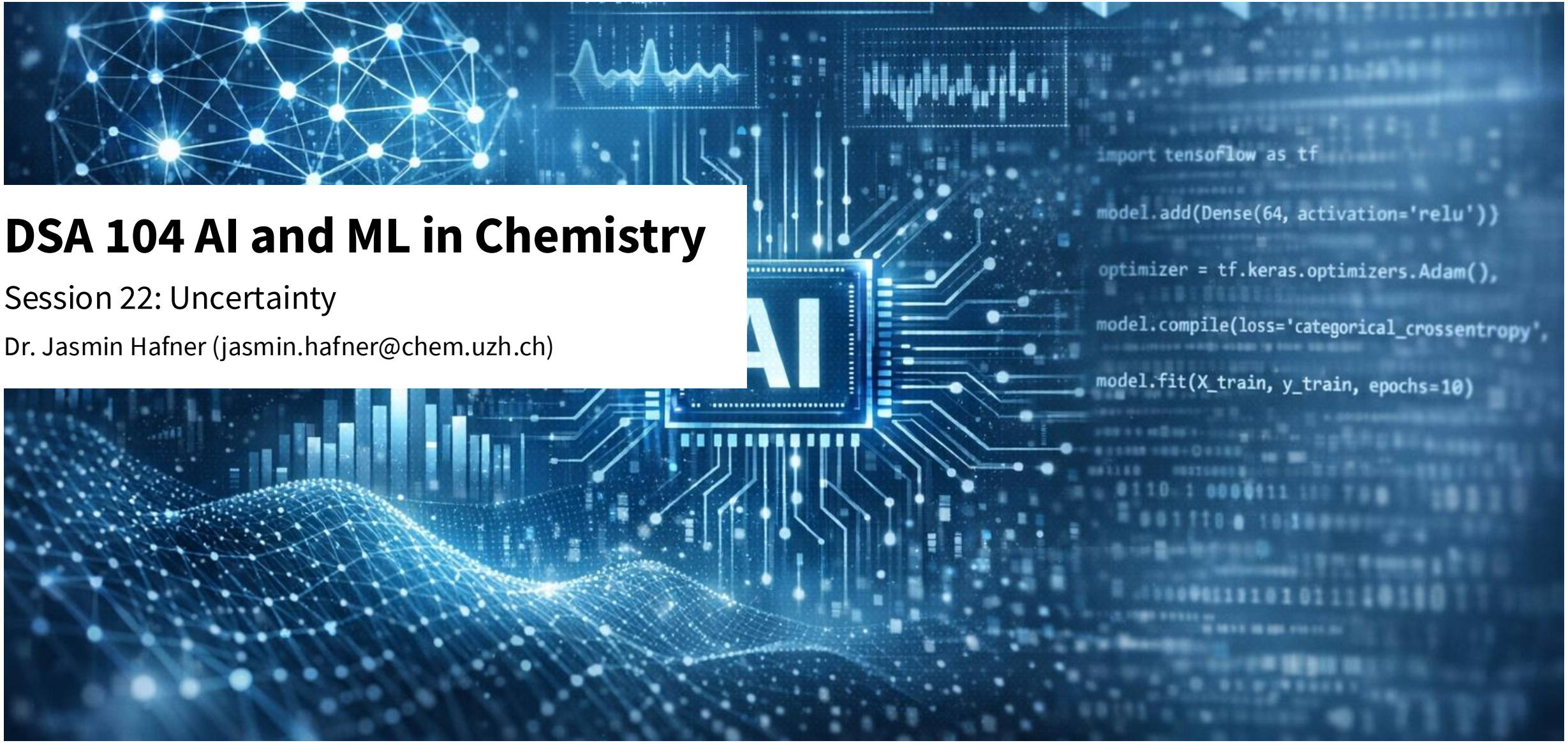
Department of Chemistry



DSA 104 AI and ML in Chemistry

Session 22: Uncertainty

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Lesson outline

Prediction uncertainty

Data uncertainty *vs* model uncertainty

Data uncertainty in ML

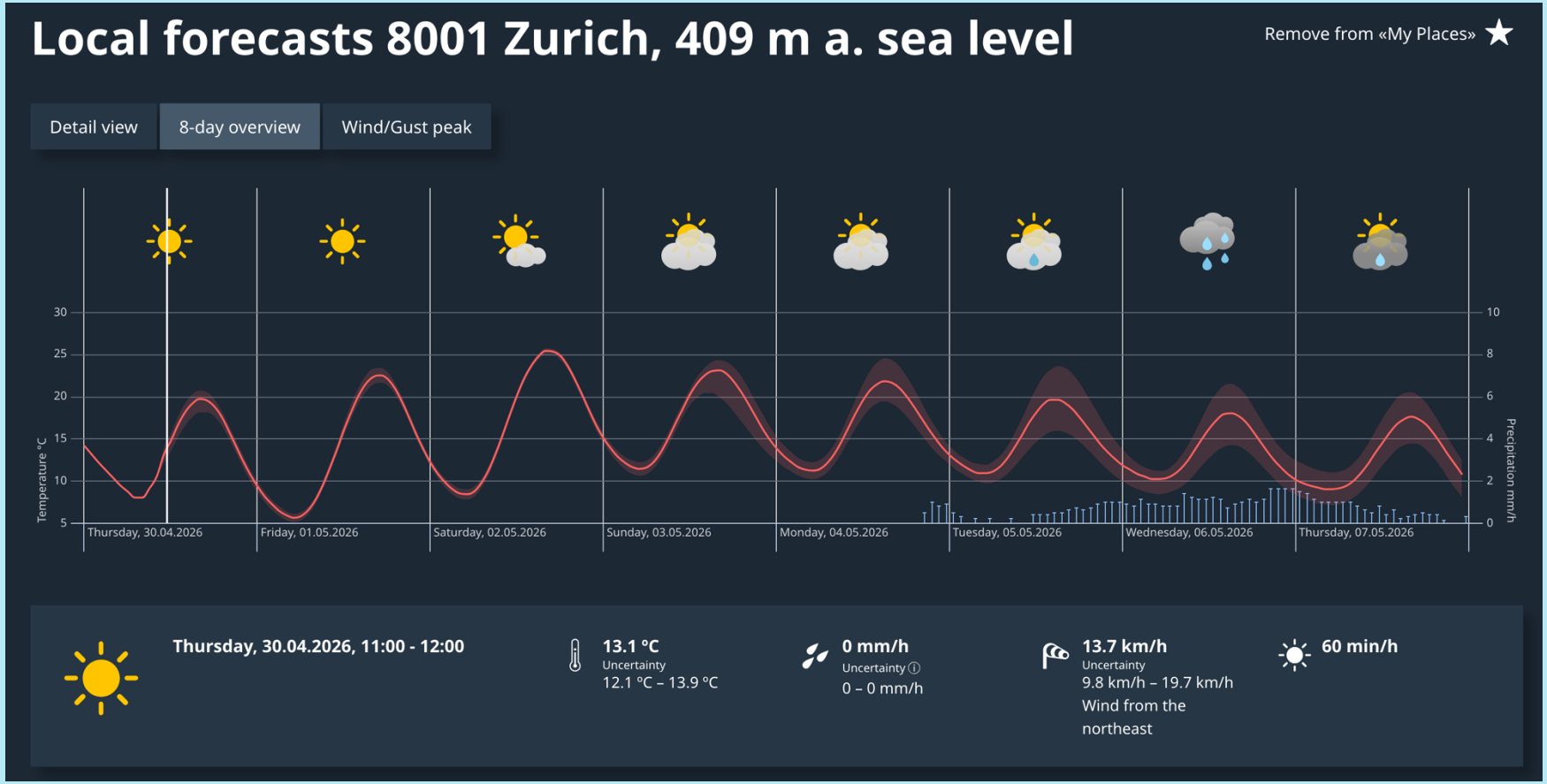
Applications

Discussion: Where do we encounter uncertainty?

Why is uncertainty here important?

Can we trust the provided uncertainties?

How do we evaluate uncertainties?



Prediction uncertainty can be as important as the prediction itself!

https://www.meteoswiss.admin.ch/local-forecasts/zurich/8001.html, consulted on 30/04/2026 at 11am

Recap AD – When (not) to trust a model prediction?

- We have seen before: Applicability Domain, but **AD ≠ model uncertainty** !

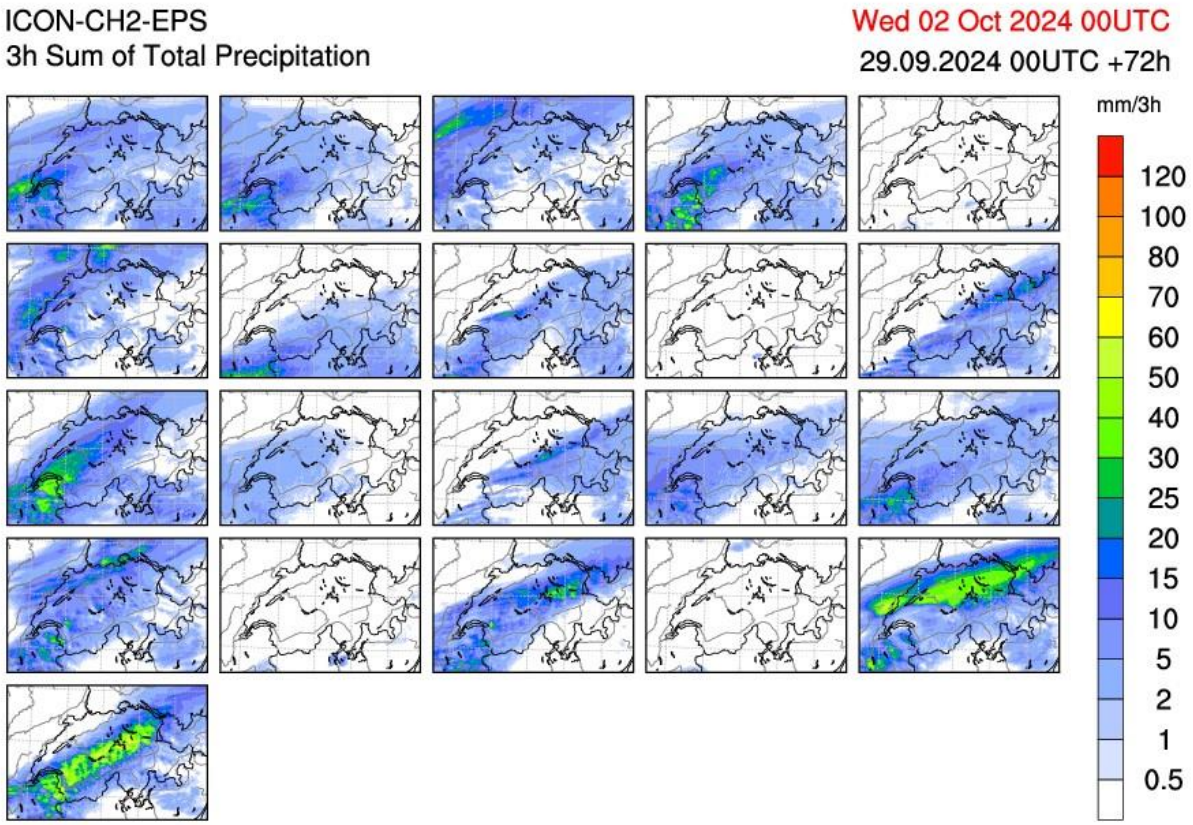
- **Step 1:** Am I within the **Applicability Domain** of the model? → Model definition / training data / AD metrics

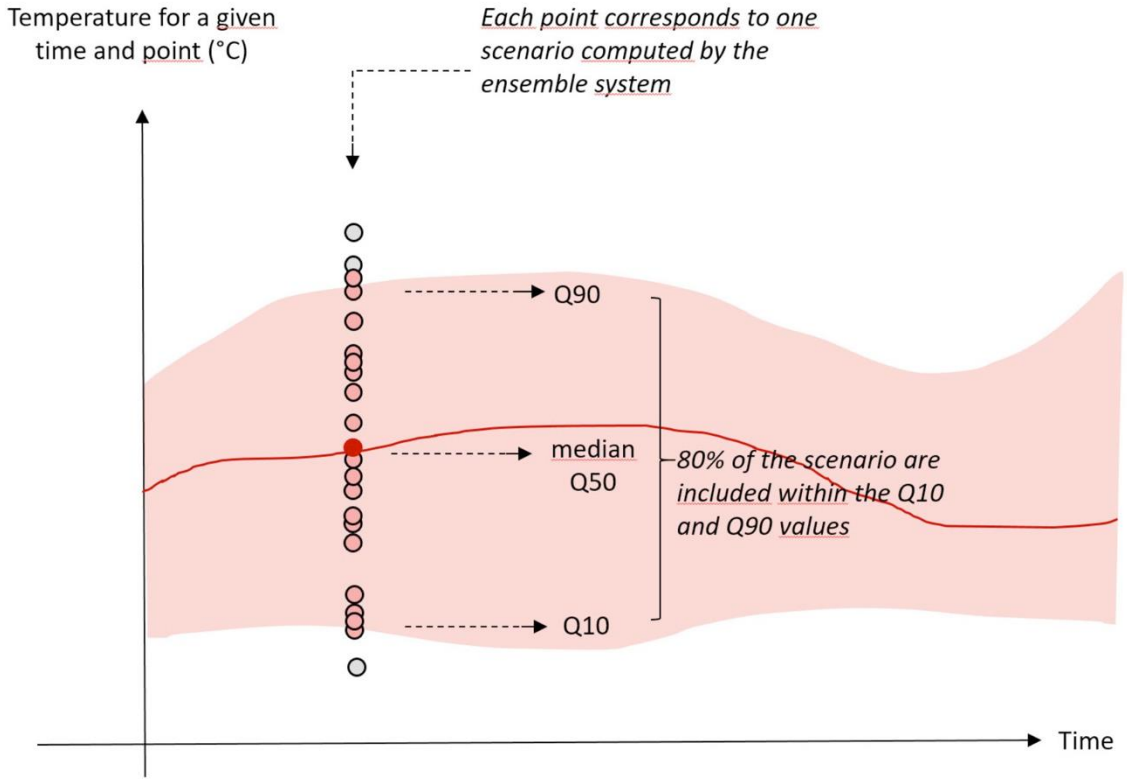
  Q: What would be the AD of the MeteoSwiss forecasting model?

If yes:
- **Step 2:** Can I **trust** the model **prediction**? → **Uncertainty** of the prediction

How can we obtain a prediction uncertainty?

Intuitive approach: Ask the same question to several models → **ensemble approach**





https://www.meteoswiss.admin.ch/weather/weather-and-climate-from-a-to-z/probability-in-forecasts.html

Exercise: Building an ensemble of models to assess prediction uncertainty

Goal:
Build an ensemble of models to predict logP with uncertainty as a standard deviation across model predictions

1) Train different regressors to predict logP on the ESOL

2) Predict logP values for our test molecules (from lecture on AD) with each model

3) Obtain uncertainty estimations for each compound from the ensemble predictions

Discuss:

- Are the results in line with the AD metrics from last week?

- Which predictions would you trust?

Where does prediction uncertainty come from?

Dissecting uncertainty: Data uncertainty *VS* model uncertainty

DATA uncertainty (aleatoric)

Is the data used for the prediction to be truseted?

Sources:

- Inherent noise in observations
- Experimental variability

→ *Irreducible error*

MODEL uncertainty (epistemic)

Is the prediction supported by data?

Sources:

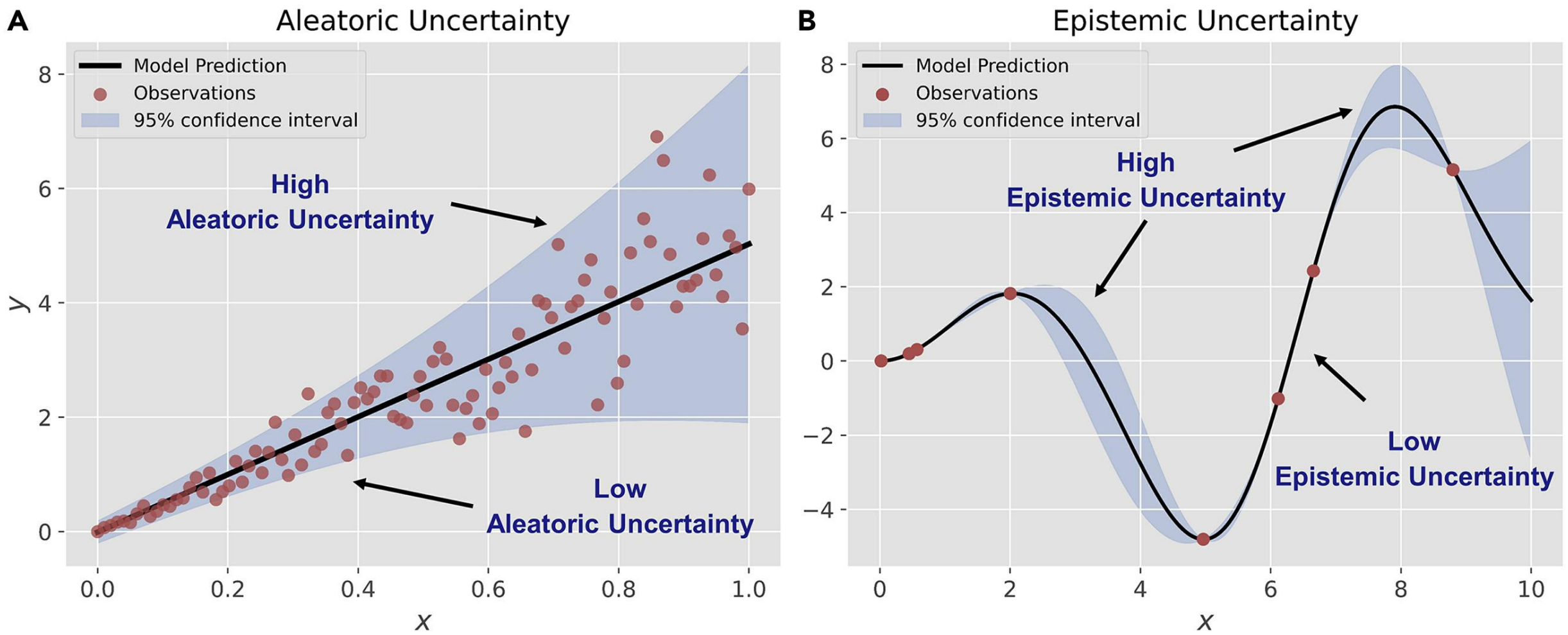
- Lack of knowledge / missing data
- Simplifying model assumptions
- Bias in knowledge / data/ measurements

→ *Reducible error (add data, refine features, optimize model architecture/parameters, …)*

Total PREDICTIVE uncertainty

https://ntrs.nasa.gov/api/citations/20230005909/downloads/DATAWorks_UQ_mini-tutorial_2023%20v2.pdf

Data uncertainty *VS* model uncertainty



Yu J, Wang D, Zheng M, **Uncertainty quantification: Can we trust artificial intelligence in drug discovery?** Science, 2022; 25

Data uncertainty

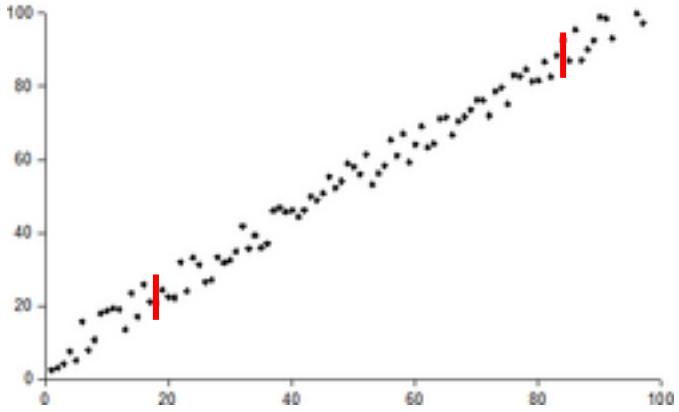
Uncertainty propagation: Uncertainty in the data affects uncertainty of the predicted target variable
→ manage your expectations!

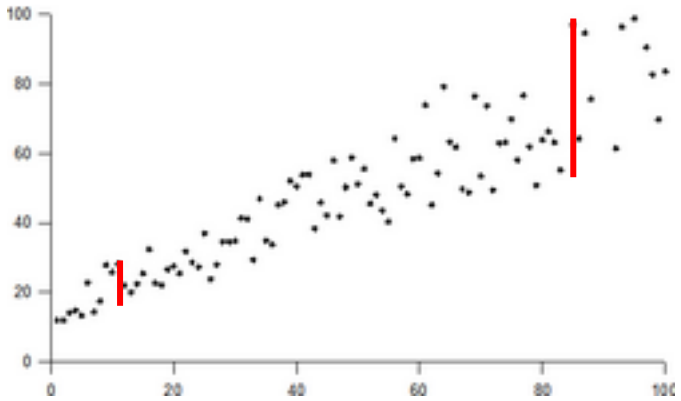
Examples of **high-uncertainty data** in chemistry:

- Toxicity assays (e.g., LD_{50} in different organisms)
- Environmental endpoints (e.g., biotransformation half-lives, sorption to soil (K_{oc}))
- Drug metabolism / activity (e.g., microsomal clearance rates)
- …

”if it interacts with biology, it gets messy”

Data uncertainty (noise) can be ***homoscedastic*** or ***heteroscedastic***





https://en.wikipedia.org/wiki/Homoscedasticity_and_heteroscedasticity

Uncertainty quantification

- Methods for quantifying uncertainty:
  - **Ensemble** methods:
    - Variability across separate models with different initialisation/training data (first examples/exercise)
    - Variability across trees in a Random Forest as a standard deviation
    - Quantile regression: If the target variable is *not normally* distributed
    - Monte Carlo (MC) dropout for Neural Networks: dropout during training *AND* prediction
  - **Bayesian** methods:
    - Gaussian Process (GP)
    - Bayesian Networks (BNs)
    - Bayesian Neural Networks (BNNs) : weights are *distributions* instead of fixed values
  - **Conformal prediction:** can be wrapped around any model (*model-agnostic*)

Yu J, Wang D, Zheng M, **Uncertainty quantification: Can we trust artificial intelligence in drug discovery?** Science, 2022; 25
https://slaclab.github.io/USPAS_ML/past_sessions/winter_2022/slides/Day7_Uncertainty.pdf
Bayesian neural networks: https://www.geeksforgeeks.org/deep-learning/differences-between-bayesian-networks-and-neural-networks/

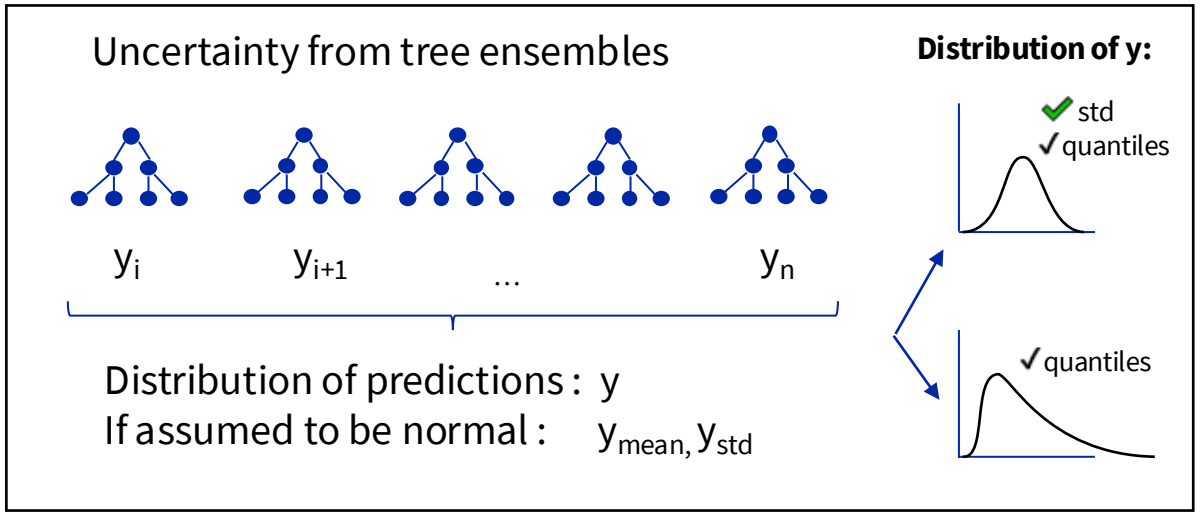
Predicting uncertainty

Let's look at two methods in detail:

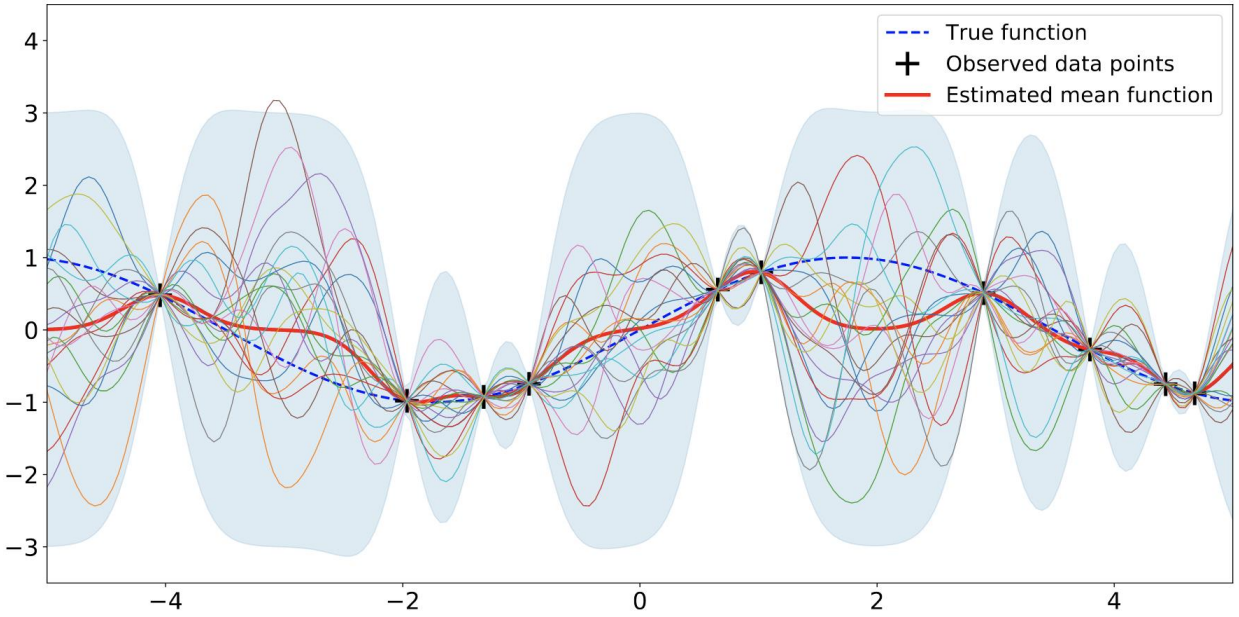
- Random Forest:
  - **Variance across single tree** predictions as a standard deviation
  - Prediction **quantiles** of single-tree predictions

- Gaussian Processes
  - **Probability distribution** over functions
  - Reports standard deviation of **output distribution** *by design*

Random Forest



Gaussian process



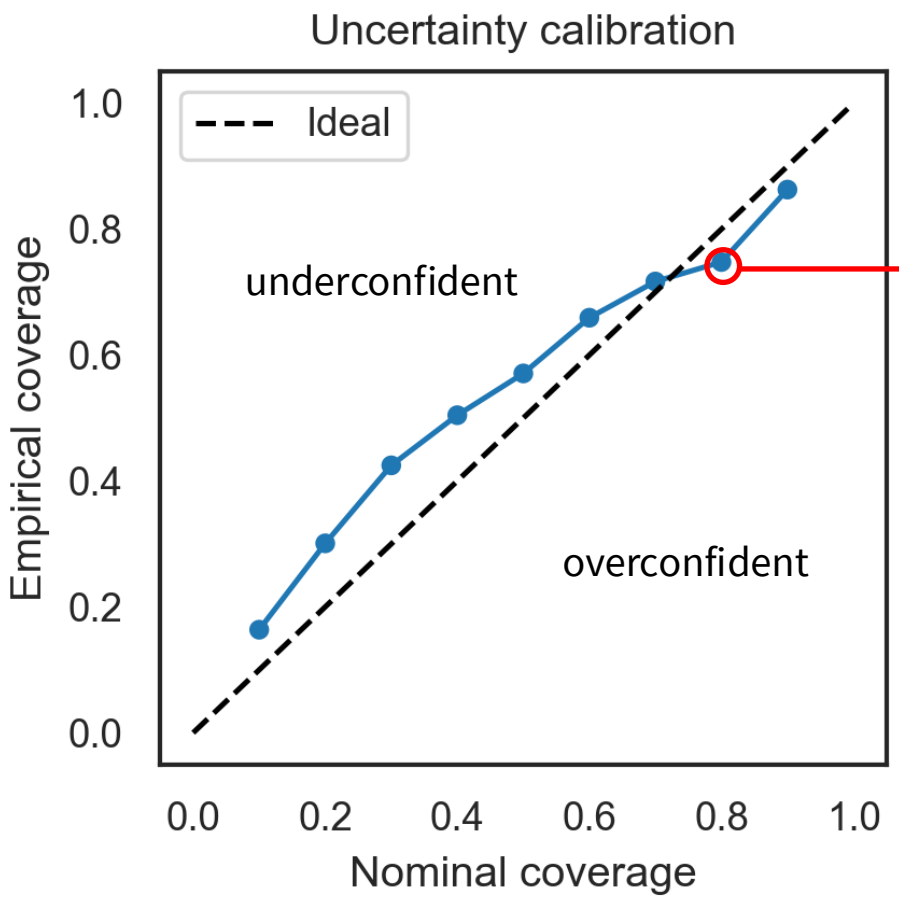
Gaussian process regressors: J. Wang, An Intuitive Tutorial to Gaussian Processes Regression, *arXiv*, 2022, preprint, arXiv:arXiv:2009.10862, DOI: 10.48550/arXiv.2009.10862.

Calibrating predictive uncertainty - Regression

When can we trust the prediction ? → When the **predicted distributions** reflect the prediction error

Perfectly calibrated model: direct match between prediction probabilities/distributions and actual success rates

E.g., When it predicts with 90% probability, it should be correct 90% of the time.

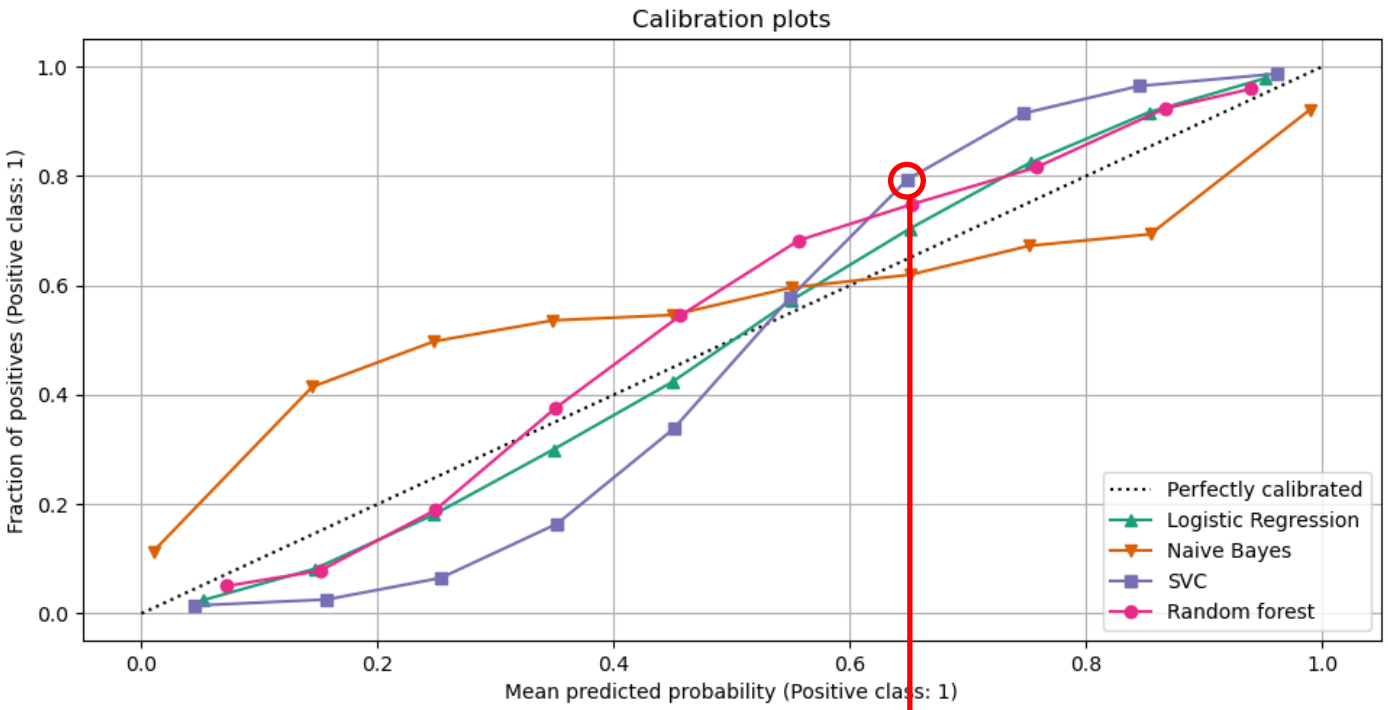


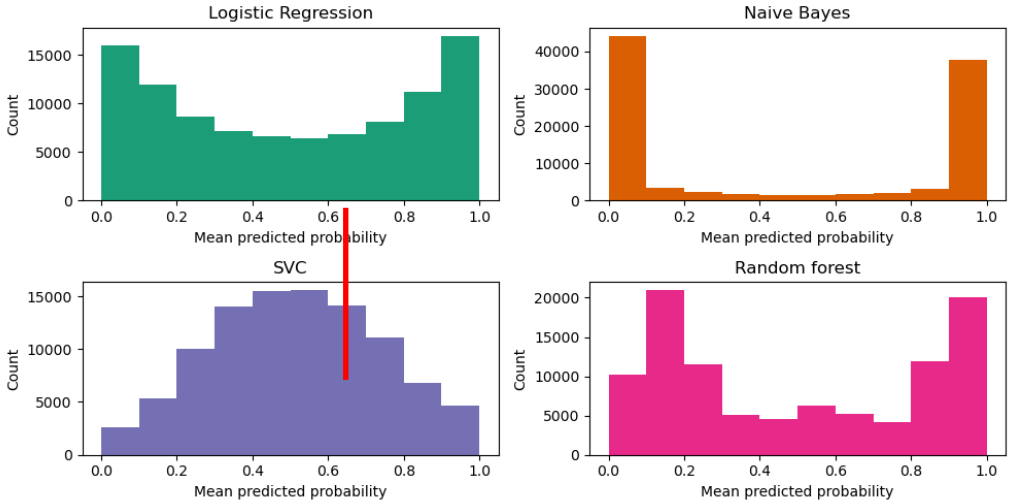
”75% of true values fall within the the 80% confidence interval of test predictions”

Calibration https://towardsdatascience.com/model-calibration-explained-a-visual-guide-with-code-examples-for-beginners-55f368bafe72/

Calibrating predictive uncertainty

When can we trust the prediction ? → When **class probabilities** reflect the prediction error





”On average predicts 65% probability, is correct 80% of time → underconfident”

Class probabilities can be obtained from most classifers in scikit-learn.
E.g., RandomForestClassifier:
`predict_proba(X)`

More on how to calibrate lassification models: https://scikit-learn.org/stable/modules/calibration.html

Exercise: Prediction uncertainty

1) Build a Random Forest to predict logP on the ESOL data using features of your choice and a
   - Random Forest Regressor
   - Gaussian Process Regressor

2) Retrieve the standard deviation for each prediction for the test set

3) Compare the predicted standard deviations with the prediction error

4) Plot the calibration curves for each model

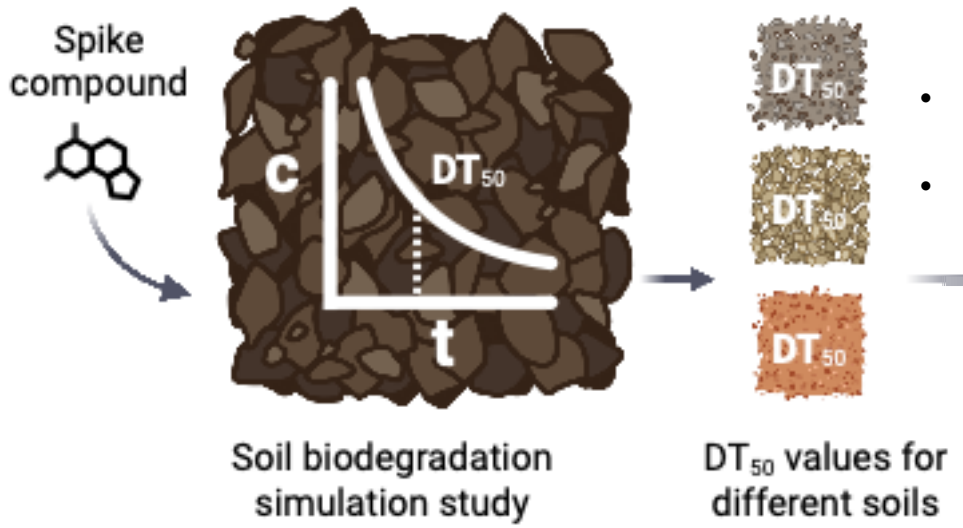
Questions:

- Which model is better calibrated?

Why uncertainty matters: Example from environmental chemistry

Objective: Predict soil biotransformation half-lives (DT_{50}) for pesticides

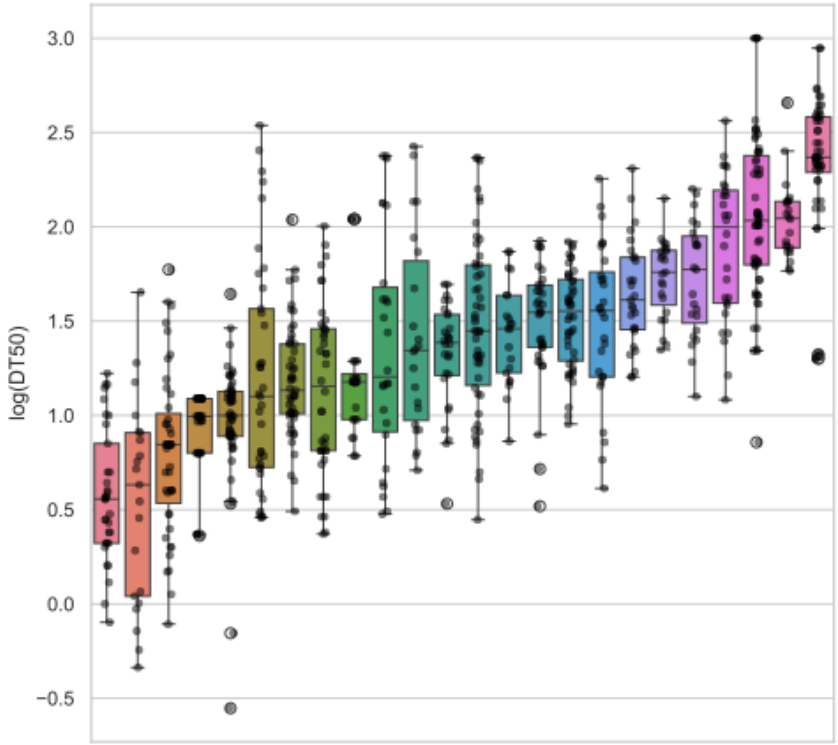
Experiment
Biotransformation half-lives observed in different soils



- 1-60 half-life values for ~900 pesticides
- One DT_{50} distribution (mean, std) per substance

→ High **aleatoric** uncertainty
→ **Heteroscedastic** noise

Data

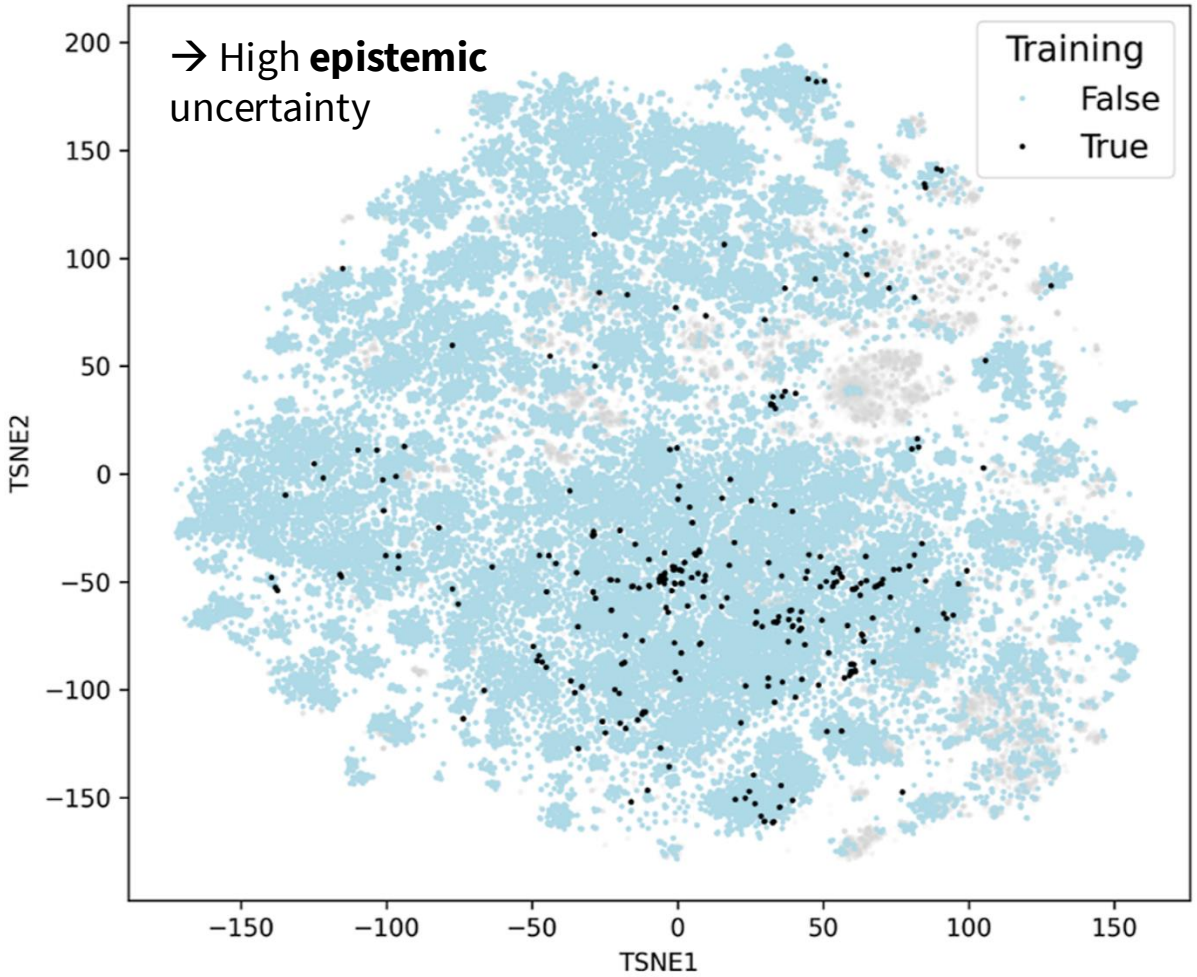


24 substances with highest variability

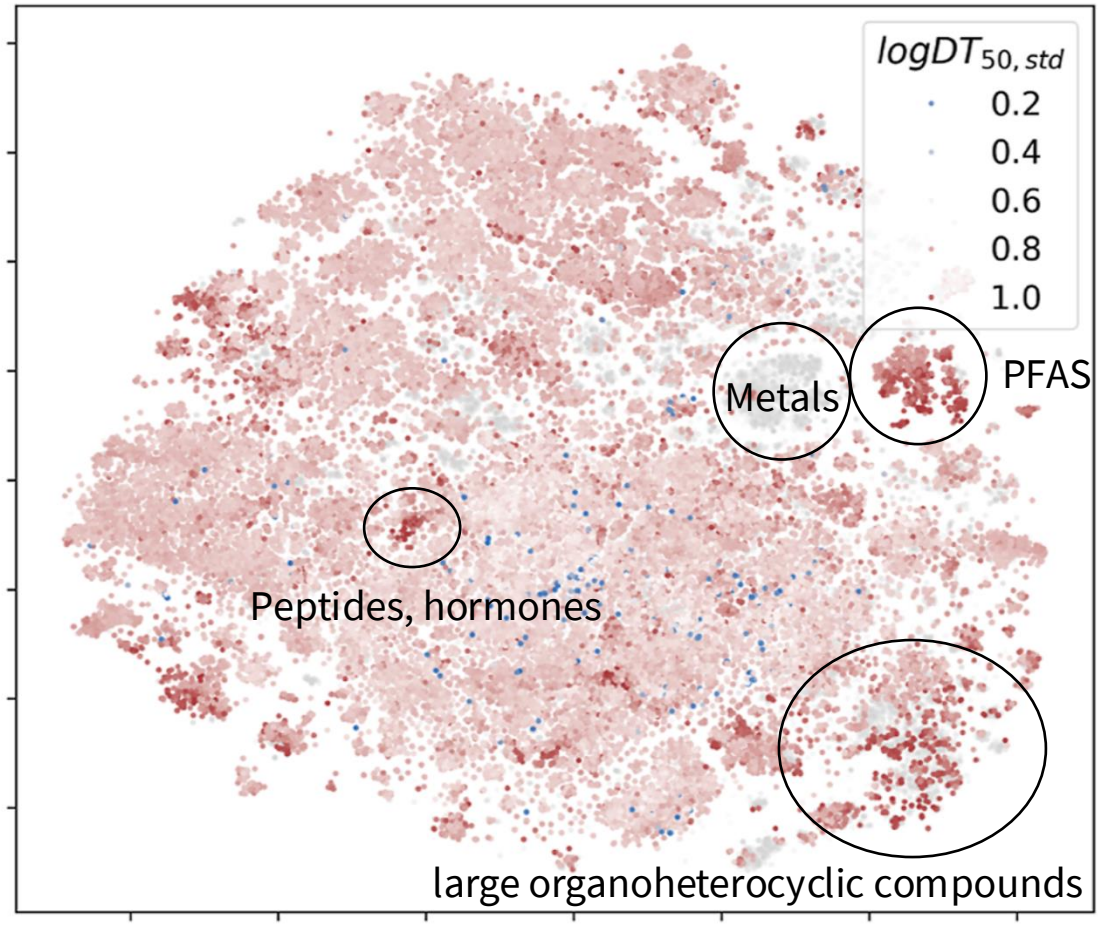
M. Salz, et al., Confidently Uncertain: Probabilistic Machine Learning to Predict Soil Biotransformation Half-Lives, *Environ. Sci. Technol.*, DOI:10.1021/acs.est.6c03516.

Why uncertainty matters: identify data gaps

Chemical space: All **marketed** chemicals
Black dots: Chemicals with **soil** biotransformation **half-lives**



Dark red: Uncertainty hot spots → data gaps!
Light grey: Inorganics / composite structres beyond AD



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Similar work: K. von Borries et al., Uncertainty-aware machine learning to predict non-cancer human toxicity for the global chemicals market, *Nat Commun*, 2026, **17**, 647.

Take-home message

- Applicability Domain (AD) → Should I **apply** the model?
- Prediction uncertainty → How **reliable** is the prediction?
- Know your **data quality** – can you trust the data points? Do you expect experimental variability?

  → ***aleatoric*** *uncertainty*

- Know your **data domain** (e.g., chemical space) – What can the model learn given the chosen model architecture and data coverage?

  → ***epistemic*** *uncertainty*

Keep in mind:

"All models are wrong, but some are useful" *George E.P. Box*

https://en.wikipedia.org/wiki/All_models_are_wrong